Aim: To study the cellular parameters network of a smartphone particularly signal strength & different cellular network

Theory:

Part 1: Evolution of Cellular Network Generations

Generation	Era / Rollout	Peak Data Speeds	Core Modulation / Technologies	Primary Use Case
2G	1990s	~100 – 400 Kbps	GSM, CDMA, GPRS, EDGE	Digital voice calling and SMS text messaging.
3G	2000s	~2 – 42 Mbps	UMTS, HSPA, HSPA+	Mobile internet browsing, basic video calling, and email.
4G	2010s	~100 – 300 Mbps	LTE, LTE-Advanced, WiMAX	High-definition video streaming, online gaming, and mobile hotspots.
5G	2020s	~1 – 20 Gbps	NR (New Radio), mmWave, Sub-6 GHz	Ultra-low latency, massive IoT, autonomous driving, and smart cities.

The fundamental theory of a cellular network relies on breaking down a vast geographic coverage area into a matrix of smaller, overlapping hexagonal spaces called cells. Each cell is powered by its own base station, or cell tower, which transmits low-power radio signals over fixed directional frequencies. By intentionally limiting the range of each tower, a network can achieve frequency reuse, allowing towers located just a few kilometers apart to operate on the exact same channel without causing destructive signal interference. This geometric distribution dramatically multiplies the capacity of the network, enabling millions of subscribers to share a finite amount of available radio spectrum simultaneously.

As a user moves while actively using their smartphone, the device continuously evaluates the network parameters. It compares the signal power and clarity from its current connection against neighboring cells. When the user crosses a boundary and the signal quality of an adjacent tower surpasses the current one, the network executes a handover (or handoff). This process instantly routes the active data stream or voice call to the new tower without dropping the connection. Behind the scenes, these towers are tied back to a centralized Core Network. This core serves as the brain of the operation, managing subscriber authentication, tracking real-time device locations across the grid, routing internet traffic, and seamlessly bridging mobile connections to the global public telephone system.

Latency and Timing Parameters

- **Latency (Ping):** The time it takes for a data packet to travel from your smartphone to a server and back. Measured in milliseconds (ms), lower latency translates to instant responsiveness. 4G networks average **30–50 ms,** while 5G cuts this down to **1–10 ms,** which is vital for real-time applications like cloud gaming and remote surgery.
- **Jitter:** The variance or fluctuation in packet arrival times over a network connection. High jitter causes voice and video packets to arrive out of order, leading to stuttering video calls or choppy audio, even if your overall download speed is high.
- **RTT (Round Trip Time):** The total time required for a network data request to go from the smartphone to the destination and return a confirmation. While latency measures a oneway trip, RTT tracks the complete two-way handshake loop.

Capacity and Efficiency Parameters

- **Spectral Efficiency:** A metric measuring how effectively a network transfers data over a given wireless bandwidth. Measured in **bits per second per Hertz (bps/Hz),** higher spectral efficiency means the carrier can cram more data into the same slice of the radio spectrum.
- **Throughput:** The actual volume of successful data successfully delivered over the network within a specific timeframe (typically measured in Mbps or Gbps). While "bandwidth" is the maximum theoretical capacity of the link, throughput is the actual speed your phone experiences.
- **Network Density:** The number of simultaneous devices a network can support per square kilometre. 4G networks can handle roughly 100,000 devices per km^2, whereas 5G expands this capacity up to **1 million devices per km^2,** effectively eliminating network congestion in crowded stadiums or dense cities.

1G: The Brick Phone Era (1980s)

1G introduced the world to wireless calling using old-school **analogue radio signals.**

- **What it did:** It let you make wireless voice calls on the move.
- **The catches:** The sound was full of static, calls dropped constantly, and eavesdroppers could easily intercept your chats with basic radio scanners. The phones were literal bricks with terrible battery life.

2G: The Text Message Revolution (1990s)

2G flipped the switch from analogue to **digital signals,** bringing us GSM and CDMA standards.

- **What it did:** It made phone calls sound crisp, added encryption so conversations stayed private, and introduced **SMS texting.** Later tweaks even let you send picture messages and check basic emails.
- **The catches:** Internet speeds were painfully slow (under 400 Kbps), meaning you could only load text-only web pages if you were patient.

3G: The Birth of Mobile Apps (2000s)

3G officially brought **broadband internet** into our pockets.

- **What it did:** Speeds climbed up to 42 Mbps, which completely changed how we used phones. It gave birth to app stores, mobile web browsing, GPS Google Maps, and early video calls.
- **The catches:** As everyone bought smartphones, the network choked up. It simply didn't have the muscle for smooth HD video streaming or online mobile gaming.

4G: The Era of Video & Streaming (2010s)

4G rebuilt the network entirely around **high-speed internet data (IP),** powered by LTE.

- **What it did:** Speeds rocketed up to 1 Gbps, and delays dropped drastically. It made HD streaming, Uber, TikTok, and online mobile gaming a seamless part of daily life. Even normal voice calls upgraded to high-definition audio (VoLTE).
- **The catches:** It couldn't handle crowds. If you went to a packed music concert or a sports stadium, your internet bars would freeze and stop working.

5G: The Hyper-Connected World (2020s)

5G uses an ultra-flexible **"New Radio" architecture** that taps into blazing-fast high-frequency waves.

Advantages:

- **Massive Capacity:** Because towers are small and scattered, those just a few kilometres apart can reuse the exact same radio frequencies. This stops the airwaves from getting crowded, letting millions of people stay online at the same time.
- **Zero-Drop Travel:** Clever "handover" tech acts like a relay race. As you drive or ride a train, your phone seamlessly passes your call or internet connection to the next tower in milliseconds without you ever noticing.
- **Longer Battery Life:** Since a cell tower is almost always nearby, your phone doesn't have to strain its battery to blast a signal miles away. This keeps your phone running cooler and lasting longer.